

(Ass - I)

Unit - I - MATHEMATICAL Logic

1. Find out the solution with table.

(i) $P \wedge \neg Q (\neg P \vee \neg Q)$

(ii) $P \vee Q (Q \wedge \neg P)$

2. Define

(i) Notation and there types.

(ii) Statement

3. What is Tautology and explain with one example also Contradiction.

4. Find out which one is tautologies or Contradiction

(i) $(P \rightarrow (P \vee Q))$

(ii) $((P \rightarrow (\neg P)) \rightarrow (\neg P))$

(iii) $((P \rightarrow (Q \rightarrow R)) \rightarrow (P \rightarrow Q) \rightarrow (P \rightarrow R))$

(iv) $(P \wedge Q) \wedge \neg (P \vee Q)$

5. Find out the truth table where $(p \wedge r) \rightarrow p$ is a Tautology.

$$(p \wedge r) \rightarrow p$$

6. Prove that (i) $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$

$$(ii) p \rightarrow (q \rightarrow r) \Leftrightarrow p \rightarrow (\neg q \vee r)$$

7. Use a truth table to verify the De Morgan's law

$$\neg(p \vee q) \equiv \neg p \wedge \neg q.$$

8. Show that the propositions $\neg(p \wedge q)$ and $\neg p \vee \neg q$ are logically equivalent.

9. Find out the following argument is Valid $p \rightarrow q, q \rightarrow r, \therefore p \rightarrow r$

10. Obtain the disjunctive normal form of the following

$$(i) p \wedge (p \Rightarrow q) \quad (ii) \neg(p \vee q) \Leftrightarrow (p \wedge q)$$